NOTICE
OF VALIDATION

MIL-D-46209C(AR)
NOTICE 1
12 May 1987

MILITARY SPECIFICATION

DETONATOR, STAB: M44E1
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

MIL-D-46209C(AR), dated 21 April 1975, has been reviewed and determined to be valid for use in acquisition.

Preparing activity:
Army - AR

AMSC N/A FSC 1375

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MIL-D-46219B (PA)
13 April 1976
SUPERSEDING
MIL-D-46219A (MU)
28 April 1965 and
MIL-D-46218A (MU)
28 April 1965
(See 6.5)

MILITARY SPECIFICATION

DETONATOR, STAB, M22
PARTS FOR, LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for loading, assembling and packing for one type of stab detonator designated as Detonator, Stab, M22.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-650 - Explosives: Sampling Inspection and Testing

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

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DRAWINGS

PICATINNY ARSENAL

*PRODUCT AND PACKAGE DRAWINGS

7548238 - Detonator, Stab, M22, Assembly
8795797 - Disc, Detonator
8797036 - Disc Detonator Closing
8797745 - Cup, Detonator
*9258182 - Box, Set-up, Packing Ammunition, Inner for Small Explosive Components
*9258183 - Box, Set-up, Packing Ammunition, Outer for Small Explosive Components
*9258184 - Box, Packing, Ammunition for Small Explosive Components

*SPECIAL INSPECTION EQUIPMENT DRAWINGS

8816367 - Guide, Firing Pin
8816368 - Disc, Lead
8827828 - Sleeve
8827934 - Adapter, Ball
9200012 - Firing Pin
9200390 - Firing Assembly
9200602 - Bushing

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*(For meaning of asterisk (*) see paragraph 6.4)

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and Assemblies.-The components and assemblies shall comply with all requirements specified on drawing (dwg.) 7548238, and with all requirements specified in applicable specifications.

3.3 Moisture Content of Lead Azide.-The moisture content of the lead azide used in loading the detonator shall not exceed 0.50 percent when tested as specified in 4.5.2.

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3.4 Moisture Content of Primer Mixture.-The moisture content of the primer mixture used in loading the detonator shall not exceed 0.30 percent when tested as specified in 4.5.3.

3.5 Functioning

3.5.1 Sensitivity.-The detonator shall function on the first attempt and shall cause perforation of a 0.1345 inch thick lead disc, to a diameter of not less than 0.125 inch when tested as specified in 4.5.5.1.

3.5.2 Waterproofness.-The detonator shall comply with the requirements of 3.5.1 after being submerged in water to a minimum depth of 2 inches for not less than 48 hours, at a temperature of 70 degrees F. plus or minus 10 degrees, when tested as specified in 4.5.5.2.

*3.6 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

*3.7 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility For Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

*4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

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4.3 First Article Inspection

*4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer. The first article sample shall consist of fifty (50) complete sets of parts and sub-assemblies and 655 detonator assemblies (i.e. fifty (50) of each and every component). The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

*4.3.2 Inspections to be Performed.-See MIL-A-48078, and Table I specified herein.

4.3.3 Rejection.-(See MIL-A-48078).

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46219B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Detonator, Stab, M22, Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Detonator Assembly (Dwg. 7548238)				
	Examination for Defects	50		3.2	4.4.2.2
	Sensitivity	315/315	(a)	3.5.1	4.5.5.1/(b)
	Waterproofness	25		3.5.2	4.5.5.2/(b)
	Cup (Dwg. 8797745)				
	Examination for Defects	50		3.2	4.4.2.3
	Disc (Dwg. 8795797)				
	Examination for Defects	50		3.2	4.4.2.1
	Disc, Closing (Dwg. 8797036)				
	Examination for Defects	50		3.2	4.4.2.2

NOTES:
(a) unique sampling plan
(b) see Special Inspection Equipment Drawings

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4.4 Quality Conformance Inspection

*4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

c. Tetryl from not more than one lot.

d. Primer mixture from one batch or blending of batches which results in homogeneity of all primer mixture used in loading the entire detonator lot.

4.4.2 Examination.-(See MIL-A-48078).

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8795797
*4.4.2.1	Disc, Detonator				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of Poor Workmanship		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.2	Disc, Detonator Closing	1 OF 1	8797036
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%		Gage
102	Thickness		0.40%		Gage
Minor					
201	Evidence of Poor Workmanship		0.65%		Visual

NOTES:

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8 3722

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8797745
*4.4.2.3	Cup, Detonator				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup, cracked or split		100.0%		Visual
Major					
101	Diameter of hole, minimum		1.0%		Gage
102	Concentricity of flash hole with inside diameter		1.0%		Gage
103	Thickness through bottom		1.0%		Gage
104	Length		1.0%		Gage
105	Outside diameter		1.0%		Gage
Minor					
201	Inside diameter		1.0%		Gage
202	Evidence of poor workmanship		1.0%		Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.4	Detonator, M22, Loading Assembly Prior to Loading	1 OF 2	7548238
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Coating compound or lacquer on exposed surface of disc		0.40%	3.1	Visual
102	Binder-Lubricant content of tetryl	(a)		3.1	4.5.1
103	Moisture content of lead azide	(a)		3.3	4.5.2
104	Moisture content of primer mixture	(a)		3.4	4.5.3
105	Composition of primer mixture	(a)		3.1	4.5.4/4.5.4.1/ 4.5.4.2/4.5.4.3/ 4.5.4.4
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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10 3722

QUALITY CONFORMATION INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
*4.4.2.4	Detonator, M22, Loading Assembly Prior to Loading				7548238
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
NOTES:(a)	The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one (1) sample of each eight (8) hours of detonator production. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, that quantity of material represented by the sample shall be rejected. If detonators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample, together with all detonators loaded with the nonconforming material, shall be rejected. The tests shall be performed as specified in 4.5.1, 4.5.2, or 4.5.3 as applicable.				

NOTES:

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3923

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46219B (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Detonator, M22, Loading Assembly, After Loading		7548238
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup cracked or split		100.0%	3.1	Visual
Major					
101	Length		0.40%	3.2	Gage
102	Diameter		0.40%	3.2	Gage
103	Disc loose, punctured, torn or wrinkled to such a degree that the contents are exposed		0.40%	3.2	Visual
104	Color of lacquer incorrect at either end		0.40%	3.2	Visual
105	Cup damaged except as described in critical above		0.40%	3.2	Visual
106	Crimp not a full 360 degrees		0.40%	3.2	Visual
107	Sensitivity	315/315	(a)	3.5.1	4.5.5.1/(b)
108	Waterproofness	25		3.5.2	4.5.5.2/(b)
109	Lacquer missing or incomplete		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:
(a) unique sampling plan
(b) see Special Inspection Equipment Drawings

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12 3232

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.6	Box, Set-Up, Inner, After Loading and Prior to Closing Slide Cover		9258182
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Filler and cushion assembly missing		100.0%	3.1	Visual
2	Filler and cushion assembly improperly assembled		100.0%	3.1	Visual
3	Stiffner missing		100.0%	3.1	Visual
4	Slide cover assembled reversed		100.0%	3.1	Visual
Major					
101	Release paper missing		0.40%	3.1	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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13 3222

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.7	Box, Set-Up, Outer, with Inner Set-Up Boxes Assembled and prior to Assembling Top				9258183
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Red surface on inner set-up boxes do not alternate on same side of outer set-up boxes		100.0%	3.1	Visual
2	Stiffner not facing one direction		100.0%	3.1	Visual
Major A					
101	Inner box does not fit properly in outer box		0.40%	3.1	Visual
102	Fillers missing		0.40%	3.1	Visual
Major B					
131	Label not securely applied to box		0.40%	3.1	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.1	Visual
202	Box improperly taped		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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14 3332

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.8	Sealed Packed Barrier Bag, Prior to Packing in Box	1 OF 1	9258183
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Waterproofness of barrier bag destroyed by rupture, puncture or separation heat seal		0.40%	3.1	Visual
102	Evidence of crushed box		0.40%	3.1	Visual
NOTE:	Package should be opened if there is evidence of crushing				
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.1	Visual
202	Marking of barrier bag misleading or unidentifiable		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.9	Box, Packing, Ammunition Prior to Assembling Cover and Strapping	1 OF 1	9258184
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Bags containing set-up boxes damaged		0.40%	3.1	Visual
102	Cushioning material incomplete or missing		0.40%	3.1	Visual
103	Marking missing, misleading or unidentifiable		0.40%	3.1	Visual
104	Bottom cushioning missing or of incorrect thickness		0.40%	3.1	Visual
Minor					
201	Contents loose within box		0.65%	3.1	Visual-Manual
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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16 3222

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46219B (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.10	Box, Packing, Ammunition	1 OF 1	9258184
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Strapping missing, broken, loose or improperly fastened		0.40%	3.1	Visual-Manual
102	Box damaged		0.40%	3.1	Visual
103	Marking missing, incomplete or illegible		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the First Article or Quality Conformance Inspection tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test Methods and Procedures

4.5.1 Determination of Binder-Lubricant Content of Tetryl.-The binder-lubricant content of the tetryl shall be determined for each batch or blend of batches at the time of loading. A sample of twenty (20) grams shall be selected and tested as specified in 4.5.1.1. If the binder-lubricant content is in excess of the percentage specified, the lot of tetryl shall be rejected and all detonators loaded with questionable tetryl shall be rejected.

4.5.1.1 An accurately weighed portion of five (5) grams shall be transferred to a tared medium porosity sintered glass crucible. The sample shall be extracted with four 10 milliliter (ml.) portions of acetone, maintaining each portion in contact with the residue for approximately five (5) minutes before applying suction. Any lumps which may persist shall be broken up and the sides of the crucible washed down with each 10 ml. portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 95 degrees Centigrade plus or minus 5 degrees C. for 1/2 hour, cooled in a desiccator and weighed. The above extraction shall be repeated, drying and weighing with individual 10 ml portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The percent of binder-lubricant content of tetryl shall be calculated as follows:

$$\text{Percent total binder-lubricant} = \frac{A-B}{W}(100)$$

A = Weight of crucible and sample before extraction
B = Weight of crucible and sample after extraction
W = Weight of the sample

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4.5.2 Moisture Content of Lead Azide.-The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample in sufficient quantity to perform the test shall be selected and tested as specified in 4.5.2.1. If the moisture content is in excess of the percent specified in 3.3 the lot shall be rejected and all detonators loaded with the questionable lead azide shall be rejected.

4.5.2.1 The moisture content of the lead azide shall be determined in accordance with MIL-STD-650, Method 101.5 except, use a two gram sample and cover the dish with a sheet of poly-tetrafluoroethylene.

4.5.3 Moisture Content of Primer Mixture.-The moisture content of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample of 3 grams shall be selected and tested in 4.5.3.1. If the moisture content is in excess of the percentage specified in 3.4, the batch or blend of batches of primer mixture shall be rejected and all detonators loaded with the questionable primer mixture shall be rejected.

4.5.3.1 An accurately weighed 0.7 gram portion of the sample shall be dried to constant weight in an oven, maintained at 60 degrees C. plus or minus 5 degrees C., or in a desiccator, containing calcium chloride. The percent moisture of the primer mixture shall be calculated as follows:

$$\text{Percent moisture} = \frac{A-B}{W}(100)$$

A = Weight of sample before drying
B = Weight of sample after drying
W = Weight of sample

4.5.4 Composition of Primer Mixture.-The composition of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample of 4 grams shall be selected and tested as specified in 4.5.4.1, 4.5.4.2, 4.5.4.3 and 4.5.4.4. If the composition of the primer mixture fails to conform to the requirements specified on the applicable drawing, the batch or blend of batches of primer mixture shall be rejected and all detonators loaded with the questionable primer mixture shall be rejected.

4.5.4.1 Potassium Chlorate.-Approximately 1 gram of the primer mixture shall be accurately weighed and transferred to a tared ignited filtering crucible of approximately 15 ml. capacity. Three ml. of lead azide-saturated distilled water shall

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be added at a temperature of 25 degrees C. plus or minus 2 degrees C., agitated for exactly 1 minute, and if necessary, any lumps shall be broken up with a glass rod, taking care not to disturb the asbestos mat. Suction shall be applied. This procedure shall be repeated five times, making a total of six ml. extractions. The sample shall be rinsed in the filtering crucible with a few ml. of alcohol and ether; dried for approximately 15 minutes at 95 degrees C. plus or minus 2 degrees C., cooled in a desiccator, and weighed. The percent of potassium chlorate should be calculated as follows:

$$\text{Percent chlorate} = \frac{A-B}{W}(100)$$

A = Weight of crucible and sample before extraction
B = Weight of crucible and sample after extraction
W = Weight of sample

4.5.4.2 Lead Azide.-The residue remaining from the potassium chlorate shall be extracted with 5 ml. portions of saturated ammonium acetate solution at about 25 degrees C. (hot saturated ammonium acetate solution must not be used as it dissolves antimony sulfide to some extent) each portion being agitated for about 30 seconds, until the washings no longer give a yellow precipitate with a few drops of potassium dichromate solution. The use of more than 60 ml. of ammonium acetate solution should be avoided. The residue shall be washed with distilled water, rinsed with a few ml. of alcohol and ether, dried for approximately 15 minutes at 95 degrees C. plus or minus 5 degrees C., cooled in a desiccator and weighed. The percent of lead azide should be calculated as follows:

$$\text{Percent lead azide} = \frac{A-B}{W}(100)$$

A = Weight of resuidue before extraction
B = Weight of residue after extraction
W = Weight of sample

4.5.4.3 Antimony Sulfide.-The residue remaining from the lead azide determination shall be extracted with concentrated hydrochloric acid until free from antimony sulfide (this extraction can be made without removing the sample from the filtering crucible and thus running the risk of disturbing the asbestos mat). Suction shall be applied and most of the antimony sulfide extracted with cool concentrated hydrochloric acid, then rinsed thoroughly with hot concentrated hydrochloric acid. The crucible and residue shall be rinsed thoroughly, ignited to remove separated sulfur and organic matter, cooled in a desiccator and weighed. The percent of antimony sulfide shall be calculated as follows:

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Percent of antimony sulfide = $\frac{A-B}{W}$ (100)

A = Weight of residue before extraction
B = Weight of residue after extraction
W = Weight of sample

4.5.4.4 Carborundum.-The original weight of the ignited filtering crucible shall be subtracted from the weight of the crucible and residue after the antimony sulfide determination. The percent of carborundum shall be calculated as follows:

Percent carborundum = A-B (100)

A = Weight of the crucible and residue from the antimony sulfide
B = Weight of the original sample

4.5.5 Functioning

4.5.5.1 Sensitivity.-A sample of three hundred and fifteen (315) detonators shall be randomly selected from each lot. If three (3) or more detonators fail to comply with the specified requirements, the lot shall be rejected. If one or two defectives occur, a second sample of 315 detonators shall be selected and tested. If, in the combined first and second sample, a total of four (4) or more defectives occur, the lot shall be rejected. The test shall be performed as specified herein using test equipment as specified in 4.4.4. The detonator shall be assembled to the applicable test fixture specified in 4.4.4 so that the firing pin will strike the sensitive end of the detonator. A steel ball 1 3/16 inch in diameter and weighing 3.95 ounces maximum, shall be dropped on the firing pin to function the detonator from a height of 12 inches maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

4.5.5.2 Waterproofness.-A sample of 25 detonators shall be selected at random from each lot. If one or more detonators fail to comply with the requirement specified the lot shall be rejected. The test shall be performed as specified herein using test equipment as specified in 4.4.4. The detonator shall be immersed under water as specified for the time specified. Within one hour after removal from the water the detonator shall be subjected to the sensitivity test specified in 4.5.5.1. Any detonator which fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

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4.5.5.3 Check Test for Possible Deterioration.-The following test is designed to prevent the loading of detonators, which have deteriorated, into a major ammunition item. The test shall be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in that contract. The contracting officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the check test for possible deterioration. A sample of 315 detonators shall be selected from each lot. If four (4) or more detonators fail to comply with the requirement specified the lot shall be rejected. The test shall be performed as specified in 4.5.5.1 using test equipment specified in 4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

*5.1.1 Level C.-The detonators shall be packaged in accordance with dwgs. 9258182 and 9258183.

*5.2 Packing

*5.2.1 Level C.-The detonators shall be packed as specified on dwg. 9258184.

*5.3 Marking.-Detonator containers and bags shall be marked in accordance with instructions contained in dwgs. 9258182, 9258183 and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering Data.-See MIL-A-48078 (PA)

6.2 Inspection Equipment Designs.-See MIL-A-48078 (PA).

6.3 Submission of Inspection Equipment Designs for Approval.-See MIL-A-48978. Submit equipment designs as required to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T, Dover, New Jersey 07801.

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6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate content irrespective of the marginal notations and relationship to the last previous issue.

6.5 Supersession Data.-This specification includes the requirements of MIL-D-46218A (MU) dated 28 APRIL 1965.

Custodian:
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Preparing Activity:
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OMB Approval
No. 22-R255

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